

Anchor-Pure Reconstruction of Regular Tetration: Riemann–Hilbert Selection, Analytic Base Flow, and Blind Validation

Mixed-Base Tetration — Paper VI

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Abstract

We develop an anchor-pure method for reconstructing the regular Kneser tetration T_d at a target base $d > e^{1/e}$ from a single known anchor T_e . No tetration, superlogarithm, hub, or response-kernel data at the target base are used in the construction. The key representation is

$$T_\beta = Q_\beta \circ U_\beta, \quad U_\beta = \text{id} + p_\beta,$$

where Q_β is a raw branch obtained by stopped logarithmic peeling and p_β is a periodic degree-one correction. We derive the exact base derivative of Q_β , prove cancellation of the nonperiodic tails in the hub equation, select the base-response field by a Riemann–Hilbert problem, and obtain a closed periodic evolution on a fixed weighted analytic path space. We also develop a finite certification framework based on contraction bounds, segment composition, two-sided fixed-point gluing, and cut-plane continuation.

Two independent numerical architectures validate the reconstruction at base 2. A 65-node global base flow reconstructs T_2 with a blind pointwise discrepancy of 4.0×10^{-21} on 48 off-grid points and resolves the periodic hub to about 30.6 decimal digits. An independently implemented ten-segment re-anchoring march agrees with both the first reconstruction and an independently computed target-base reference to 7.6×10^{-11} . The observed failure and successful subdivision of the final edge segment agree with the predicted Ovsyannikov step-size regime. These experiments establish a reproducible reconstruction method and validate its internal error predictors. The accompanying certification framework is rigorous, while completion of the full interval march is left as an open computer-assisted problem.

1 Introduction

Regular tetration to a base b is a function satisfying

$$T_b(x+1) = b^{T_b(x)}, \quad T_b(0) = 1. \tag{1}$$

For $b > e^{1/e}$, the regular real branch is selected by its complex Kneser geometry rather than by the real functional equation alone. Kneser’s construction provides the classical holomorphic solution at the natural base, while later uniqueness and continuation results characterize regular branches for wider classes of bases [7, 8, 9, 10]. High-precision evaluation of such a branch is usually performed independently at each base. This paper asks a different question:

*See the Authorship and AI Contribution Statement at the end of this manuscript: the research reported here was carried out in a directed human–AI mode, with the author acting as research director and substantial parts of the mathematical development and of the numerical work produced by AI systems under his direction.

Can a regular tetration at a new base be reconstructed from one already known regular tetration and the target base alone?

We call a construction *anchor-pure* when every target state is generated from the anchor and the base parameter, without importing target-base tetration, superlogarithm, or response-kernel values.

The central decomposition is

$$T_\beta = Q_\beta \circ U_\beta, \quad U_\beta = \text{id} + p_\beta, \quad (2)$$

where $\beta = \log \log b_\beta$. The raw branch Q_β is obtained from a previously reconstructed anchor by logarithmic peeling. It already satisfies the target functional equation but has the wrong height normalization. The correction U_β is a degree-one lift, so p_β is periodic and can be treated in a weighted Wiener space. This separates the nonperiodic tetration from the periodic state that is actually evolved.

Differentiation in the base parameter produces a cohomological equation for the Abel coordinate. The homogeneous ambiguity is removed by a Riemann–Hilbert selection condition at the conjugate complex fixed points. After an exact cancellation between the base derivative of the raw branch and the logarithmic tail of the response field, the evolution closes on the periodic hub. The resulting vector field is integrated along the base parameter by short re-anchored segments.

Main results

The analytic contribution consists of five parts.

1. We formulate the reconstruction in compatible periodic and nonperiodic analytic spaces and derive the normalized QH representation (2).
2. We construct and prove uniqueness of the Riemann–Hilbert response field in the stated Hardy class, including explicit Lipschitz and analytic-tail bounds.
3. We prove the exact raw-branch identity

$$\partial_\beta Q_\beta = -Q'_\beta S_{Q,\beta}$$

and the resulting periodic hub equation in which all nonperiodic tails cancel.

4. We establish local analytic continuation in the base parameter on one fixed weighted path space, together with a posteriori contraction bounds, Gronwall propagation, segment composition, and path-consistency gates.
5. We give a finite certification theorem: interval RH records, negative linear radii polynomials, endpoint inclusions, two-sided fixed-point gluing, and cut-plane continuation imply identification of the reconstructed family with regular Kneser tetration.

The numerical contribution is a blind cross-validation at $d = 2$. The first method interpolates a family of RH-selected kernels in the base coordinate and integrates a global Picard–Chebyshev flow. The second recomputes the RH field locally and repeatedly re-anchors the raw branch. Their agreement provides a nontrivial ring closure because the two methods use different kernel sampling, evolution, and normalization architectures.

The complete interval proof of the concrete march is not assumed in the numerical claims. Section 12 reports the blind experiments, while Appendix A records the current interval components and the remaining computer-assisted obligations.

Relation to Papers I–V

This paper is the sixth of the mixed-base tetration series and consumes, rather than re-derives, several of its earlier results. Its notation was developed independently and does not agree symbol for symbol with the earlier papers, so we record the correspondence explicitly.

Papers I and II [1, 2] construct the mixed-base phase $\Phi_{b,d}(\theta) = \lim_{n \rightarrow \infty} (\text{slog}_b(T_d(n+\theta)) - (n+\theta))$, prove the groupoid law $F_{b,d} = F_{b,c} \circ F_{c,d}$ and the exact mean antisymmetry $\mu_{b,d} = -\mu_{d,b}$, and show that the lifted phase maps $\tilde{H}_{b,d} = \text{id} + \Phi_{b,d}$ are orientation-preserving circle diffeomorphisms with $\tilde{H}_{b,d}^{-1} = \tilde{H}_{d,b}$. Paper III [3] removes the real-channel conditionality of that construction by a stopped logarithmic chain; the stopped peeling of Section 8 below is the same scheme, used there to generate the raw branch rather than to evaluate a phase. Paper IV [4] supplies the exact hub calculus $\mu_{b,d} = m(d) - m(b) + \Delta$ with $m(b) = \mu_{e,b}$, the measured hub column, and the first-order response kernel. Paper V [5] formulates the selection principle for the base derivative $V = \partial_\beta A_b$, proves its uniqueness, and solves the associated scalar Riemann–Hilbert problem constructively.

The dependence on Paper V is the closest. The geometric field Γ_β of Section 7 and its cocycle (10) are the field $G = \partial_\beta \chi / \chi'$ and the cocycle of [5, Lemma 7]; the boundary data of the selection problem, its Dini regularity at the punctures, and the conjugate-function solution are established there [5, Lemmas 7–8, Thm. 11], and the Hardy correction in (12) below is that solution written in the present normalization. What is new here is not the pointwise selection principle but its use as the vector field of a flow in the base parameter acting on a periodic state, together with the anchor-pure raw branch, the exact tail cancellation of Theorem 8.5, and the certification framework of Section 11.

Table 1: Notation dictionary between this paper and Papers I–V.

here	Papers I–V	object
Γ_β	G [5]	Koenigs geometric field $\partial_\beta \chi / \chi'$
$v_{\text{geo}}^\pm$	$V_{\text{geo}} \circ T_d$ [5]	geometric height field
$V_\beta = \partial_\beta A_\beta$	V, V_d [4, 5]	base derivative of the Abel coordinate
S_β	S_d [4]	tower tail $\sum_{j \geq 0} T(\cdot + j) / T'(\cdot + j)$
$G_\beta[h]$	$-G_d$ [4]	periodic response kernel
p_β	$\Phi_{c,b}$ [1, 2]	periodic hub state / mixed-base phase
$U_\beta = \text{id} + p_\beta$	$\tilde{H}_{c,b}$ [2]	lifted phase map
$\mu = \int_0^1 p_\beta$	$\mu_{c,b} = m(b) - m(c) + \Delta$ [3, 4]	mean shift

Two conventions differ and are retained as they stand, each being natural in its own paper. First, the periodic response field carries opposite signs: $G_\beta = \mathcal{R}_\beta^h - S_\beta$ in (13) below, whereas $G_d = S_d - V_d \circ T_d$ in [4, Thm. 7.3], so that $G_\beta = -G_d$ under the identification of Table 1. Second, [5] writes G for the Koenigs field, which is Γ_β here, and reserves no symbol for the periodic kernel. Remark 8.9 records the identification of the evolved state p_β with the phase of Papers I–IV, together with the numerical cross-checks it makes available.

Organization. Sections 2–4 state the branch assumptions and establish the a posteriori and propagation estimates. Sections 5–7 introduce the analytic spaces, normalized QH state, and RH selection operator. Sections 8–10 derive the periodic base flow and Kneser identification. Section 11 gives the finite certification framework, and Section 12 presents the numerical reconstructions. Sections 13–14 discuss the consequences, open problems, and conclusion. The appendices collect the interval precertificate and the computational implementation, certificate format, and reproducibility safeguards.

2 Preliminaries and branch assumptions

For $b > e^{1/e}$ set

$$\beta = \log \log b, \quad b_\beta = \exp(e^\beta), \quad E_\beta(y) = b_\beta^y = \exp(ye^\beta).$$

Then

$$\partial_\beta E_\beta(y) = yE'_\beta(y). \quad (3)$$

Let T_β denote the branch to be reconstructed, normalized by $T_\beta(0) = 1$, and let $A_\beta = T_\beta^{-1}$. Put

$$V_\beta = \partial_\beta A_\beta, \quad v_\beta = V_\beta \circ T_\beta, \quad S_\beta(x) = \sum_{j \geq 0} \frac{T_\beta(x+j)}{T'_\beta(x+j)}.$$

Differentiating the Abel equation gives

$$V_\beta(E_\beta(y)) - V_\beta(y) = -yA'_\beta(y), \quad (4)$$

$$v_\beta(x+1) - v_\beta(x) = -\frac{T_\beta(x)}{T'_\beta(x)}. \quad (5)$$

Differentiating $T_\beta(A_\beta(y)) = y$ yields the exact transport equation

$$\partial_\beta T_\beta(x) = -T'_\beta(x)v_\beta(x). \quad (6)$$

Thus T is constant along the characteristics $dx/d\beta = v_\beta(x)$.

Hypothesis 2.1 (K1: common Kneser geometry). On each compact base interval $I \Subset (-1, \infty)$, the regular branch T_β^K extends holomorphically to a common strip, is real on $\mathbb{R}$, has $(T_\beta^K)' > 0$ on $[0, 1]$, and its forward images stay a positive distance from all logarithmic cuts used in the construction.

Hypothesis 2.2 (K2: Koenigs–theta structure). At the primary complex fixed-point pair $L_\beta, \bar{L}_\beta$ of E_β , the Abel coordinate has the form

$$A_\beta^K = \chi_\beta + \vartheta_\beta \circ \chi_\beta, \quad \vartheta_\beta(w+1) = \vartheta_\beta(w),$$

with $\vartheta'_\beta(w) \rightarrow 0$ at the corresponding translation ends.

Hypotheses K1–K2 encode the classical complex geometry of the regular branch.

Remark 2.3 (Historical base-holomorphy route). One may additionally assume that $\beta \mapsto A_\beta^K$ is C^1 on common domains when comparing directly with a pre-existing Kneser family. This auxiliary assumption is not used in the principal identification theorem: base regularity is obtained from the constructed Volterra flow after pointwise static uniqueness has identified each reconstructed branch.

3 A posteriori fixed-point certificates

Let $(X, \|\cdot\|)$ be a Banach space.

Theorem 3.1 (Contraction certificate). *Let $\mathcal{F} : \bar{B}_R(\tilde{H}) \rightarrow X$ be Lipschitz with constant $q < 1$, and let $r = \tilde{H} - \mathcal{F}(\tilde{H})$. If $\|r\| \leq (1-q)R$, then $\mathcal{F}$ has a unique fixed point H_* in the ball and*

$$\|H_* - \tilde{H}\| \leq \frac{\|r\|}{1-q}.$$

Even without the self-map inequality, there is at most one fixed point in the ball and every such fixed point satisfies the same error bound.

Proof. For H in the ball,

$$\|\mathcal{F}(H) - \tilde{H}\| \leq qR + \|r\| \leq R.$$

The Banach fixed-point theorem gives existence and uniqueness. The a posteriori estimate follows from

$$H_* - \tilde{H} = \mathcal{F}(H_*) - \mathcal{F}(\tilde{H}) - r.$$

□

Theorem 3.2 (Newton–Kantorovich radii polynomial). *Let $F : U \subset X \rightarrow X$ be Fréchet differentiable on a neighborhood of $\overline{B}_r(\tilde{H})$ and let $B \in \mathcal{L}(X)$ be injective. Assume*

$$\|BF(\tilde{H})\| \leq Y, \quad \|I - BDF(\tilde{H})\| \leq Z_0,$$

and

$$\|B(DF(H) - DF(\tilde{H}))\| \leq K \|H - \tilde{H}\| \quad (H \in \overline{B}_r(\tilde{H})).$$

If

$$p(r) = Y + (Z_0 - 1)r + Kr^2 < 0,$$

then F has a unique zero H_* in the ball and

$$\|H_* - \tilde{H}\| \leq \frac{Y}{1 - Z_0 - Kr} \leq r.$$

Proof. The Newton-like map $\mathcal{N}(H) = H - BF(H)$ has derivative norm at most $Z_0 + Kr < 1$. The radii inequality implies that $\mathcal{N}$ maps the ball into itself. Apply Theorem 3.1. □

Remark 3.3. The theorem is retained as a general tool. Its quadratic profile is not used for the segment proof below unless an implementation independently validates all three same-space hypotheses by interval arithmetic; see Remark 9.4.

4 Error propagation and path consistency

Lemma 4.1 (Integral Gronwall inequality). *If $\phi, \alpha : [a, b] \rightarrow [0, \infty)$ are continuous and*

$$\phi(t) \leq \phi(a) + \int_a^t (M\phi(s) + \alpha(s)) ds,$$

then

$$\phi(t) \leq e^{M(t-a)}\phi(a) + \int_a^t e^{M(t-s)}\alpha(s) ds.$$

Theorem 4.2 (A posteriori characteristic-flow bound). *Let G_β be uniformly Lipschitz in x with constant M_1 . Let $X' = -G_\beta(X)$ and let $\tilde{X}$ be a C^1 approximation with residual*

$$R(\beta) = \tilde{X}'(\beta) + G_\beta(\tilde{X}(\beta)).$$

Then

$$|X(\beta_b) - \tilde{X}(\beta_b)| \leq e^{M_1|\Delta\beta|} \left(|X(\beta_a) - \tilde{X}(\beta_a)| + \int_{\beta_a}^{\beta_b} |R(s)| ds \right).$$

The same statement holds for the full field v .

Lemma 4.3 (Lipschitz constant of an exact segment flow). *For two exact solutions starting at x, y ,*

$$|X(\beta_b) - Y(\beta_b)| \leq e^{M_1|\Delta\beta|} |x - y|.$$

If G is C^1 in x , then the derivative $J = \partial_x X$ solves

$$J' = -\partial_x G_\beta(X)J, \quad e^{-M_1|\Delta\beta|} \leq J \leq e^{M_1|\Delta\beta|}.$$

Theorem 4.4 (Segment composition). *Let S_j be exact segment maps with $\text{Lip}(S_j) \leq L_j$ and let $\tilde{S}_j$ satisfy $\sup_x \|\tilde{S}_j(x) - S_j(x)\| \leq \varepsilon_j$. Then*

$$\sup_x \|\tilde{S}_m \circ \cdots \circ \tilde{S}_1(x) - S_m \circ \cdots \circ S_1(x)\| \leq \sum_{j=1}^m \varepsilon_j \prod_{k=j+1}^m L_k.$$

Theorem 4.5 (Semigroup and path-independence gate). *The exact flow satisfies*

$$S_{\beta_c, \beta_a} = S_{\beta_c, \beta_b} \circ S_{\beta_b, \beta_a}.$$

If direct and split numerical routes have certified errors ε_D and ε_S , then

$$\|\tilde{S}_D - \tilde{S}_S\| \leq \varepsilon_D + \varepsilon_S.$$

Consequently, if the measured route discrepancy exceeds this sum, at least one certificate, convention, or implementation is wrong.

5 Analytic function spaces

The estimates below use standard properties of Hardy, Wiener, and Fourier spaces; see, for example, [11, 12].

5.1 Weighted periodic Wiener spaces

Definition 5.1. For $\rho > 0$, define

$$\mathcal{A}_\rho = \left\{ f(z) = \sum_{k \in \mathbb{Z}} \hat{f}_k e^{2\pi i k z} : \|f\|_\rho = \sum_{k \in \mathbb{Z}} |\hat{f}_k| e^{2\pi \rho |k|} < \infty \right\}.$$

Let $\mathcal{A}_\rho^+$ be the closed Hardy subspace with $\hat{f}_k = 0$ for $k < 0$. We use the reflection $f^*(z) = \overline{f(\bar{z})}$ and the Hilbert-transform convention

$$\mathcal{H}(e^{2\pi i k x}) = -i \operatorname{sgn}(k) e^{2\pi i k x}, \quad \operatorname{sgn}(0) = 0.$$

Thus $\mathcal{C}_+ = \mathcal{H} - iI$ annihilates all negative Fourier modes.

Proposition 5.2. $\mathcal{A}_\rho$ is a Banach algebra of functions holomorphic in $|\Im z| < \rho$, with

$$\sup_{|\Im z| \leq \rho} |f(z)| \leq \|f\|_\rho, \quad \|fg\|_\rho \leq \|f\|_\rho \|g\|_\rho.$$

For $0 < \rho' < \rho$,

$$\|f'\|_{\rho'} \leq \frac{1}{e(\rho - \rho')} \|f\|_\rho.$$

The periodic Hilbert transform, Fourier projections, real and imaginary parts on $\mathbb{R}$, and point evaluation are bounded operators.

Remark 5.3 (Correct periodic state). A tetraton cannot belong to $\mathcal{A}_\rho$, because $T(0) = 1$ and $T(1) = b \neq 1$. The periodic objects are the lift correction $p = U - \text{id}$, the response field G , and the homogeneous RH correction.

5.2 Nonperiodic fundamental-domain spaces

For $\sigma > 0$, set

$$\Omega_\sigma = \{z \in \mathbb{C} : \text{dist}(z, [0, 1]) < \sigma\}, \quad \mathcal{B}_\sigma = H^\infty(\Omega_\sigma).$$

For $0 < \sigma' < \sigma$,

$$\|f'\|_{\mathcal{B}_{\sigma'}} \leq \frac{1}{\sigma - \sigma'} \|f\|_{\mathcal{B}_\sigma}. \quad (7)$$

Lemma 5.4 (Holomorphic composition). *If $f, g \in \mathcal{B}_\sigma$ take values in a compact set $K \Subset \Omega$ and F is holomorphic on Ω , then*

$$\|F \circ f - F \circ g\|_{\mathcal{B}_\sigma} \leq \sup_{K_\Delta} |F'| \|f - g\|_{\mathcal{B}_\sigma},$$

where K_Δ contains the line segments joining corresponding values of f and g .

Lemma 5.5 (Periodization bridge). *If a holomorphic function on Ω_σ is 1-periodic wherever both z and $z + 1$ lie in the domain, then it extends uniquely to the strip S_σ . For $0 < \rho < \sigma$,*

$$\|f\|_\rho \leq C_{\text{per}}(\sigma - \rho) \|f\|_{\mathcal{B}_\sigma}, \quad C_{\text{per}}(a) := \frac{1 + e^{-2\pi a}}{1 - e^{-2\pi a}}.$$

Proof. Translate the germ on Ω_σ by integers. The overlap identity makes the translated germs compatible and gives a unique holomorphic periodic function on S_σ . Its Fourier coefficients obey $|\widehat{f}_k| \leq \|f\|_{\mathcal{B}_\sigma} e^{-2\pi\sigma|k|}$, first on every strictly smaller contour and then in the limit. Hence

$$\|f\|_\rho \leq \|f\|_{\mathcal{B}_\sigma} \sum_{k \in \mathbb{Z}} e^{-2\pi(\sigma-\rho)|k|} = \frac{1 + e^{-2\pi(\sigma-\rho)}}{1 - e^{-2\pi(\sigma-\rho)}} \|f\|_{\mathcal{B}_\sigma}.$$

□

Lemma 5.6 (Fourier tail). *For $f \in \mathcal{A}_\rho$ and $K \geq 0$,*

$$\sum_{|k| > K} |\widehat{f}_k| e^{2\pi\rho'|k|} \leq e^{-2\pi(\rho-\rho')(K+1)} \|f\|_\rho, \quad 0 < \rho' < \rho.$$

6 Normalized QH representation

On a re-anchoring segment, let Q_β be a raw branch, holomorphic on D_Q , real on the real channel, locally injective, with $Q'_\beta \neq 0$ and

$$Q_\beta(z + 1) = b_\beta^{Q_\beta(z)}. \quad (8)$$

Let $q_\beta = Q_\beta^{-1}(1)$ and define the closed real-symmetric normalization space

$$\mathcal{A}_{\rho,0} = \{h \in \mathcal{A}_\rho : h(0) = 0, h(\bar{z}) = \overline{h(z)}\}.$$

For $h \in \mathcal{A}_{\rho,0}$ put

$$p = q_\beta + h, \quad U_{\beta,h}(z) = z + q_\beta + h(z), \quad T_{\beta,h} = Q_\beta \circ U_{\beta,h}.$$

Then $T_{\beta,h}(0) = 1$, $U(z + 1) = U(z) + 1$, and the tetration equation is automatic:

$$T_{\beta,h}(z + 1) = E_\beta(T_{\beta,h}(z)). \quad (9)$$

Definition 6.1 (Admissible normalized QH state). For a parameter list

$$P = (\rho, \sigma, M_h, m_U, m_Q, \mu_T, \Delta_Q, \Delta_E, \delta_\lambda, J_0, \mathbf{a}, \mathbf{D}, M_\Gamma, M_{\Gamma'}),$$

with $0 < \sigma < \rho$, let $\mathcal{X}_\beta^0(P)$ be the set of $h \in \mathcal{A}_{\rho,0}$ satisfying:

(Z1) $\|h\|_\rho \leq M_h$.

(Z2) $U' = 1 + h'$ obeys $\inf_{\overline{\Omega_\sigma}} |U'| \geq m_U > 0$ and $\inf_{[0,1]} U' > 0$, the image $U(\overline{\Omega_\sigma})$ stays a distance Δ_Q inside D_Q , and $|Q'_\beta| \geq m_Q$ there.

(Z3) $T \in \mathcal{B}_\sigma$, T' is zero-free, and $\|1/T'\|_{\mathcal{B}_\sigma} \leq \mu_T$.

(Z4) The finitely many transfers $T_0 = T$, $T_{j+1} = E_\beta(T_j)$, $0 \leq j \leq J_0$, stay a distance Δ_E from every logarithmic cut and Koenigs singularity used in the construction.

(Z5) Explicit sequences $a_j > 0$, $D_j \geq 0$ satisfy

$$\inf_{\Omega_\sigma} |T_j| \geq a_j, \quad \|T_j[h] - T_j[\tilde{h}]\|_{\mathcal{B}_\sigma} \leq D_j \|h - \tilde{h}\|_\rho,$$

and the tail majorants of Theorem 7.7 converge.

(Z6) $|1 - \lambda_\beta| \geq \delta_\lambda$ and $|\lambda_\beta| \geq 1 + \delta_\lambda$.

(Z7) Upper and lower Koenigs branches $\Gamma_\beta^\pm$ are defined on certified image domains containing a common complex overlap of $T(\Omega_\sigma)$, satisfy $\Gamma^- = (\Gamma^+)^*$, obey the same cocycle, and have interval bounds $|\Gamma^\pm| \leq M_\Gamma$, $|(\Gamma^\pm)'| \leq M_{\Gamma'}$.

Lemma 6.2 (Stability of the normalized state space). *Every admissible state has a positive radius $\varepsilon_0(P)$ such that every real-symmetric perturbation $\tilde{h} \in \mathcal{A}_{\rho,0}$ with $\|h - \tilde{h}\|_\rho \leq \varepsilon_0(P)$ remains admissible, with all positive margins reduced by at most a factor two.*

Proof. The normalization is preserved exactly because every perturbation vanishes at zero. Derivatives are controlled on a smaller strip by Proposition 5.2; image sets and compositions are controlled by Lemma 5.4. Zero-freeness, cut distances, and the finitely many transfer conditions persist by compactness. Taking the minimum of the finitely many resulting radii proves the claim. $\square$

7 The Riemann–Hilbert selection operator

Let L_β be a primary complex fixed point of E_β , with multiplier $\lambda_\beta = E'_\beta(L_\beta)$. On a cut-free backward orbit, let χ_β be the Abel-normalized Koenigs coordinate,

$$\chi_\beta(E_\beta(y)) = \chi_\beta(y) + 1,$$

and define

$$\Gamma_\beta(y) = \frac{\partial_\beta \chi_\beta(y)}{\chi'_\beta(y)}.$$

Differentiation gives the exact cocycle

$$\Gamma_\beta(E_\beta(y)) = E'_\beta(y)(\Gamma_\beta(y) - y). \quad (10)$$

Equation (10) is the cocycle of [5, Lemma 7], where Γ_β is denoted G ; existence, uniqueness and a constructive graded-mesh solution of the corresponding boundary problem *at a fixed base* are established there. The present section restates the selection in the normalized QH variables and

adds the operator bounds needed for the flow. For an admissible state, define the two geometric height fields

$$v_{\text{geo}}^{\pm}[h](x) = \frac{\Gamma_{\beta}^{\pm}(T(x))}{T'(x)}, \quad v_{\text{geo}}^{-} = (v_{\text{geo}}^{+})^{*}.$$

They satisfy

$$v_{\text{geo}}^{\pm}(x+1) - v_{\text{geo}}^{\pm}(x) = -\frac{T(x)}{T'(x)}. \quad (11)$$

Lemma 7.1 (Lipschitz bound for the geometric field). *Assume*

$$\|T[h] - T[\tilde{h}]\|_{\mathcal{B}_{\sigma'}} \leq C_T \|h - \tilde{h}\|_{\rho}, \quad \|T'[h] - T'[\tilde{h}]\|_{\mathcal{B}_{\sigma'}} \leq C_{T'} \|h - \tilde{h}\|_{\rho}.$$

Then

$$\|v_{\text{geo}}^{\pm}[h] - v_{\text{geo}}^{\pm}[\tilde{h}]\|_{\mathcal{B}_{\sigma'}} \leq C_{\text{geo}} \|h - \tilde{h}\|_{\rho},$$

where

$$C_{\text{geo}} = \frac{M_{\Gamma} C_T}{m_T} + \frac{M_{\Gamma} C_{T'}}{m_T^2}, \quad m_T = \inf |T'|.$$

Lemma 7.2 (Trace theorem without circularity). *The boundary difference*

$$u_h(x) = \frac{v_{\text{geo}}^{+}[h](x) - v_{\text{geo}}^{-}[h](x)}{2i}$$

extends holomorphically and periodically to each smaller strip. For $0 < \rho_u < \sigma' < \sigma$,

$$\|u_h\|_{\rho_u} \leq C_{\text{per}}(\sigma' - \rho_u) \max_{\pm} \|v_{\text{geo}}^{\pm}[h]\|_{\mathcal{B}_{\sigma'}},$$

and

$$\|u_h - u_{\tilde{h}}\|_{\rho_u} \leq C_{\text{tr}} \|h - \tilde{h}\|_{\rho}, \quad C_{\text{tr}} = C_{\text{per}}(\sigma' - \rho_u) C_{\text{geo}}.$$

Proof. The two geometric branches have the same inhomogeneity in (11); their difference is therefore periodic. It is holomorphic on the overlap and periodizes to the smaller strip. Lemma 5.5 gives the Wiener bound. No later RH theorem is used. $\square$

Let $\mathcal{C}_+ u = \mathcal{H}u - iu$, which contains only nonnegative Fourier modes, and set

$$c[h] = -(v_{\text{geo}}^{+}[h] + \mathcal{C}_+ u_h)(0).$$

Define

$$\boxed{\mathcal{R}_{\beta}^h[h] = v_{\text{geo}}^{+}[h] + \mathcal{C}_+ u_h + c[h]}. \quad (12)$$

Theorem 7.3 (RH existence and uniqueness). *For every admissible h , the field $v = \mathcal{R}_{\beta}^h[h]$ is well defined, real on $\mathbb{R}$, normalized by $v(0) = 0$, and satisfies (5). It is the unique solution of the form*

$$v = v_{\text{geo}}^{+}[h] + w, \quad w \in \mathcal{A}_{\rho_u}^{+},$$

that is real on $\mathbb{R}$ and vanishes at zero.

Proof. The periodic correction leaves the cohomological difference unchanged and cancels the imaginary boundary value. If two solutions exist, their difference is a real Hardy function. Reality pairs positive and negative modes, while the Hardy condition removes all negative modes; hence all nonzero modes vanish, and the normalization removes the constant. $\square$

Theorem 7.4 (RH Lipschitz estimate). *On a common admissibility ball and smaller domains,*

$$\left\| \mathcal{R}_\beta^h[h] - \mathcal{R}_\beta^h[\tilde{h}] \right\|_{\mathcal{B}_{\rho_u}} \leq C_{\text{RH}}(P) \left\| h - \tilde{h} \right\|_\rho,$$

with the explicit choice

$$C_{\text{RH}} \leq (1 + C_{\text{ev}})(C_{\text{geo}} + 2C_{\text{tr}}), \quad C_{\text{ev}} = 1.$$

Proof. First control the geometric branches on $\Omega_{\sigma'}$, then the periodic trace on S_{ρ_u} , and restrict both to Ω_{ρ_u} . Only then apply the bounded Hilbert transform and point evaluation. This order avoids both a trace–RH circularity and a domain mismatch. $\square$

Definition 7.5 (Complete interval RH record). For a base box I_β and a path ball $\mathfrak{B}$, an interval RH record $\mathcal{I}_{\text{RH}}(I_\beta, \mathfrak{B})$ consists of outward-rounded enclosures for all of the following objects.

1. Interval Chebyshev–Bernstein models for Q_β, Q'_β and $T_{\beta,h}, T'_{\beta,h}$ on Ω_σ , including truncation remainders, image enclosures, and strict nonvanishing and cut-distance margins, uniformly for $(\beta, h) \in I_\beta \times \mathfrak{B}$.
2. Interval Newton enclosures for L_β, λ_β and validated stopped-peeling models for $\Gamma_\beta^\pm, (\Gamma_\beta^\pm)'$ on the two image domains. The two models must have a certified common overlap, conjugation symmetry, and interval residual/tail bounds for the cocycle (10).
3. Quotient enclosures for $v_{\text{geo}}^\pm = \Gamma_\beta^\pm(T)/T'$, using the zero exclusion for T' , and an interval proof that their difference has identical boundary values at the two ends of the fundamental interval.
4. A Fourier enclosure

$$u_h = \sum_{|k| \leq K} \mathbf{u}_k e^{2\pi i k x} + \mathbf{R}_K$$

obtained by validated quadrature or interval Chebyshev–Fourier conversion. The remainder is bounded in $\mathcal{A}_{\rho_u}$ by the explicit periodization constant of Lemma 5.5 and the analytic tail of Lemma 5.6; sampling error is included.

5. Exact multiplier action of $\mathcal{C}_+$ on the interval Fourier head, the corresponding analytic tail enclosure, interval point evaluation at zero, and the normalization enclosure for $c[h]$ in (12).
6. Enclosures of $\mathcal{R}_\beta^h[h]$, its Fréchet or Lipschitz difference bounds on the whole path ball, and every downstream tail constant used in Theorem 7.7. Directed rounding, precision, library versions, and hashes of all coefficient arrays are part of the record.

No point sample, midpoint FFT, or floating-point norm may be substituted for one of these interval enclosures.

Theorem 7.6 (Soundness of the interval RH record). *If $\mathcal{I}_{\text{RH}}(I_\beta, \mathfrak{B})$ satisfies all strict margins in Definition 7.5, then for every $(\beta, h) \in I_\beta \times \mathfrak{B}$ the exact operator (12) exists, lies in the stored interval enclosure, and satisfies the stored uniform norm and Lipschitz bounds.*

Proof. Items 1–2 validate every nonperiodic composition before division. Item 3 therefore encloses both geometric traces and proves periodic compatibility without using the RH conclusion. Lemma 5.5 and Lemma 5.6 validate the finite Fourier enclosure in item 4. The Hardy projection is diagonal, so item 5 encloses it coefficient by coefficient and adds the rigorous tail. The interval evaluation at zero fixes the only remaining constant. The algebraic proof of Theorem 7.3 then applies to every member of the boxes, while interval composition and quotient estimates give the uniform differences recorded in item 6. $\square$

7.1 The complex tail

Define

$$S_\beta[h](x) = \sum_{j \geq 0} \frac{T(x+j)}{T'(x+j)}.$$

Theorem 7.7 (Certified tail convergence). *Let the sequences a_j, D_j from (Z5) be given, put $c = \log b$, and let $M_T \geq \|T\|_{\mathcal{B}_\sigma}$. Assume on the same ball that*

$$\|T[h] - T[\tilde{h}]\|_{\mathcal{B}_\sigma} \leq C_T \|h - \tilde{h}\|_\rho, \quad \|T'[h]^{-1} - T'[\tilde{h}]^{-1}\|_{\mathcal{B}_\sigma} \leq C_{T'^{-1}} \|h - \tilde{h}\|_\rho.$$

Define

$$M_0 = M_T \mu_T, \quad M_j = \mu_T c^{-j} \prod_{r=1}^{j-1} a_r^{-1} \quad (j \geq 1).$$

If $\sum_j M_j < \infty$, then $S_\beta[h]$ converges normally on Ω_σ . If additionally

$$\sum_{j \geq 1} \frac{1}{c^j} \left[C_{T'^{-1}} \prod_{1 \leq r < j} a_r^{-1} + \mu_T \sum_{1 \leq r < j} D_r a_r^{-2} \prod_{\substack{1 \leq s < j \\ s \neq r}} a_s^{-1} \right] < \infty,$$

and the finite zeroth term obeys

$$\left\| \frac{T[h]}{T'[h]} - \frac{T[\tilde{h}]}{T'[\tilde{h}]} \right\|_{\mathcal{B}_\sigma} \leq (C_T \mu_T + M_T C_{T'^{-1}}) \|h - \tilde{h}\|_\rho,$$

then S_β is locally Lipschitz in h .

Proof. For $j \geq 1$, the functional equation gives the exact identity

$$\frac{T(x+j)}{T'(x+j)} = \frac{1}{c^j T'(x) \prod_{r=1}^{j-1} T(x+r)}.$$

The $j = 0$ summand is T/T' and is bounded separately by M_0 . The first claim follows. For two states, use the displayed zeroth-term bound, telescope the products of inverse factors for $j \geq 1$, and use $|z^{-1} - w^{-1}| \leq |z - w|/(|z||w|)$. $\square$

The periodic response field is

$$G_\beta[h] = \mathcal{R}_\beta^h[h] - S_\beta[h]. \tag{13}$$

The two terms have the same cohomological defect, so $G_\beta[h]$ is 1-periodic.

8 Raw-branch differentiation and periodic hub flow

Let Q_β be constructed by stopped logarithmic peeling from an arbitrary reconstructed anchor T_c . The seeds are

$$z_N = \alpha_\beta T_c(N+x) + c_{0,\beta}, \quad \alpha_\beta = \frac{\log c}{\log b_\beta}, \quad c_{0,\beta} = \frac{\log \alpha_\beta}{\log b_\beta}.$$

A stopped-peeling certificate SP controls the function sequence; an SP' certificate additionally controls the base derivatives of every peeling level and their summable tails.

Definition 8.1 (SP' certificate). An $SP'(K)$ certificate consists of the SP bounds on a compact set K , interval bounds $|\partial_\beta z_N| \leq Z_N$ with $Z_N/|\partial_x z_N| \rightarrow 0$, and a summable sequence $\varepsilon_N^{(\beta)}$ such that the differentiated peeled approximants obey

$$\|\partial_\beta Q_{N+1} - \partial_\beta Q_N\|_K \leq \varepsilon_N^{(\beta)}, \quad \sum_{N \geq N_0} \varepsilon_N^{(\beta)} < \infty.$$

For $y_{k+1} = \log_{b_\beta} y_k$ and $y_0 = z_N$, a sufficient interval majorant is obtained by telescoping the differentiated chains with products of inverse modulus bounds:

$$\varepsilon_N^{(\beta)} \leq C_N \prod_{k=1}^{N-J_0} m_{N,k}^{-1}, \quad \sum_{N \geq N_0} C_N \prod_{k=1}^{N-J_0} m_{N,k}^{-1} < \infty.$$

Theorem 8.2 (Exact base derivative of the raw branch). *Under $SP'(K)$, Q_β is differentiable in β on K , and*

$$\boxed{\partial_\beta Q_\beta = -Q'_\beta S_{Q,\beta}, \quad S_{Q,\beta}(x) = \sum_{j \geq 0} \frac{Q_\beta(x+j)}{Q'_\beta(x+j)}}. \quad (14)$$

Equivalently,

$$S_{Q,\beta}(x) = \frac{Q_\beta(x)}{Q'_\beta(x)} + \sum_{j \geq 1} \frac{1}{c^j Q'_\beta(x) \prod_{r=1}^{j-1} Q_\beta(x+r)}, \quad c = \log b_\beta.$$

Proof. Differentiated peeling obeys

$$\partial_\beta y_{k+1} = -y_{k+1} + \frac{\partial_\beta y_k}{y_k \log b}.$$

The SP' majorant makes the differentiated approximants uniformly Cauchy. For $W = \partial_\beta Q/Q'$, differentiation of the prelimit functional equation gives

$$W(x+1) - W(x) = \frac{Q(x)}{Q'(x)}.$$

Both W and $-S_Q$ satisfy this equation and vanish in the deep channel; their difference is periodic and tends to zero, hence vanishes identically. $\square$

Corollary 8.3 (Derivative of the normalization point). *If $Q_\beta(q_\beta) = 1$, then*

$$q'_\beta = S_{Q,\beta}(q_\beta).$$

Lemma 8.4 (Tail transformation). *For $U = \text{id} + p$ and $T = Q_\beta \circ U$,*

$$S_\beta[h](x) = \frac{S_{Q,\beta}(U(x))}{1 + p'(x)}.$$

Theorem 8.5 (Exact hub reduction). *Let $T_\beta = Q_\beta \circ (\text{id} + p_\beta)$ and let G_β be the periodic response field (13). Then the transport equation (6) is equivalent to the purely periodic evolution*

$$\boxed{\partial_\beta p_\beta = -(1 + p'_\beta)G_\beta[h_\beta]}. \quad (15)$$

All nonperiodic tail contributions cancel exactly.

Proof. By the chain rule and Theorem 8.2,

$$\partial_\beta T = -(Q' S_Q) \circ U + (Q' \circ U) \partial_\beta p.$$

Using $T' = (Q' \circ U)(1 + p')$, $\mathcal{R}^h = G + S$, and Lemma 8.4, the transport right-hand side equals

$$-(Q' \circ U)(1 + p')G - (Q' S_Q) \circ U.$$

Cancel the common tail term and divide by $Q' \circ U$. □

Theorem 8.6 (Closed normalized flow). *Writing $p_\beta = q_\beta + h_\beta$ with $h_\beta \in \mathcal{A}_{\rho,0}$ gives*

$$\boxed{\partial_\beta h_\beta = \Psi_\beta[h_\beta] := -(1 + h'_\beta)G_\beta[q_\beta + h_\beta] - q'_\beta.} \quad (16)$$

For every admissible state, $\Psi_\beta[h](0) = 0$, so the flow remains in the linear space $\mathcal{A}_{\rho,0}$.

Corollary 8.7 (Anchor gate). *Write*

$$\mu(\beta) = \int_0^1 p_\beta(x) dx, \quad \overline{G} = \int_0^1 G_\beta[h_\beta](x) dx$$

for the mean of the hub state and the mean of the periodic response field. At the anchor, where $p = 0$,

$$\partial_\beta p|_{\text{anchor}} = -G_\beta[0], \quad \left. \frac{d\mu}{d\beta} \right|_{\text{anchor}} = -\overline{G}.$$

Away from the anchor,

$$\frac{d\mu}{d\beta} = -\overline{G} - \int_0^1 p'(x)G_\beta[h_\beta](x) dx.$$

Lemma 8.8 (Re-anchoring composition). *If $U_i = \text{id} + p_i$ are degree-one lifts and the composition is defined on the chosen strips, then*

$$U_2 \circ U_1 = \text{id} + p_{21}, \quad p_{21} = p_1 + p_2 \circ (\text{id} + p_1).$$

Thus an already reconstructed state may serve as the next anchor without importing any external target-base data.

Remark 8.9 (The hub state is the mixed-base phase of Papers I–IV). The raw branch of this section is generated by stopped peeling from the current anchor T_c , that is $Q_\beta = \log_{b_\beta}^{om}(\alpha_\beta T_c(m + \cdot) + c_{0,\beta})$ in the limit. By the value-level statement of [2, Rem. 7.1], $T_b(j + \tilde{H}_{b,c}(\theta)) = \alpha T_c(j + \theta)(1 + O(1/s_j))$ with hyper-exponentially small relative error, so the stopped peeling chain converges to

$$Q_\beta = T_\beta \circ \tilde{H}_{\beta,c}$$

on the certified channel. Comparing with $T_\beta = Q_\beta \circ U_\beta$ and using $\tilde{H}_{\beta,c}^{-1} = \tilde{H}_{c,\beta}$ [2, Thm. A] identifies

$$U_\beta = \tilde{H}_{c,\beta}, \quad p_\beta = \Phi_{c,\beta}, \quad \mu(\beta) = \mu_{c,\beta} = m(\beta) - m(c) + \Delta,$$

the last equality by the hub decomposition [4, Prop. 8.1]. The periodic state evolved by (15) is therefore the mixed-base phase of Papers I–IV, and the base flow of this paper is an evolution equation for the hub function m . Two consequences are worth recording.

(i) *The anchor gate is the hub slope.* With $c = e$, Corollary 8.7 states $dm/d\beta|_{b=e} = -\overline{G}$, which under the sign convention of Table 1 is the hub slope $\overline{G}_e$ of [4, Cor. 7.4(ii)], measured there as 2.191463277066. The anchor gate of the present flow and that measurement are the same number and may be compared directly.

(ii) *A target-base check with an independent reference.* The mean of the reconstructed hub at $b = 2$ equals $\mu_{e,2} = m(2)$, which Papers III–IV determine from the stopped peel ladder to $-1.128403776628924009879217902036426267112\dots$ without any Kneser evaluation *at the target base*. Comparing the reconstructed mean against that value is thus a target-base check whose reference is produced by a different mechanism from the engine evaluation used in Section 12. This comparison was not part of the blind protocol reported below; it is recommended as an additional gate.

9 Analytic base flow on a fixed scale

For $a, \rho_0, \kappa > 0$, define the cone

$$\mathfrak{C}_{a,\kappa} = \{(t, \rho) : 0 \leq t \leq a, 0 < \rho < \rho_0 - \kappa t\}, \quad d(t, \rho) = \rho_0 - \kappa t - \rho.$$

Let $\mathfrak{C}_{a,\kappa}$ be the space of compatible paths $u(0) = 0, u(t) \in \mathcal{A}_{\rho,0}$ on the cone, with norm

$$\|u\|_{\mathfrak{C}} = \sup_{(t,\rho) \in \mathfrak{C}_{a,\kappa}} \sqrt{d(t, \rho)} \|u(t)\|_{\rho}. \quad (17)$$

This is a Banach space.

Theorem 9.1 (Ovsyannikov theorem in a fixed path space). *Assume $\kappa a < \rho_0$ and, on a scale-admissible ball,*

$$\|\hat{\Psi}_t[h]\|_{\rho'} \leq \frac{M}{\rho - \rho'}, \quad \|\hat{\Psi}_t[h] - \hat{\Psi}_t[k]\|_{\rho'} \leq \frac{L}{\rho - \rho'} \|h - k\|_{\rho}.$$

Let $\mathcal{V}f(t) = \int_0^t f(s) ds$ and

$$\mathcal{T}(u) = \mathcal{V}(\hat{\Psi}[h_a + u]).$$

Then

$$\|\mathcal{T}(u) - \mathcal{T}(v)\|_{\mathfrak{C}} \leq q \|u - v\|_{\mathfrak{C}}, \quad q = \frac{4\sqrt{2}L}{\kappa},$$

and

$$\|\mathcal{T}(0)\|_{\mathfrak{C}} \leq b_0 = 4M\sqrt{\frac{a}{\kappa}}.$$

If $q < 1$ and a ball of radius $R \geq b_0/(1 - q)$ lies entirely in the scale-admissible set, then the normalized flow has a unique solution on the segment.

Proof. Fix (t, ρ) and put $d = d(t, \rho)$. For $0 \leq s \leq t$, use the intermediate radius

$$\eta_s = \rho + \frac{d + \kappa(t - s)}{2}.$$

The scale loss and the remaining path width are both half of $d + \kappa(t - s)$. The Lipschitz integrand is therefore bounded by

$$2^{3/2}L(d + \kappa(t - s))^{-3/2} \|u - v\|_{\mathfrak{C}}.$$

Integration gives $4\sqrt{2}L/(\kappa\sqrt{d})$, and multiplication by $\sqrt{d}$ gives the contraction constant. The base term is controlled similarly using $\log(1 + x) \leq 2\sqrt{x}$. $\square$

Corollary 9.2 (A priori segment length). *There exists a κ satisfying both $4\sqrt{2}L/\kappa < 1$ and $\kappa a < \rho_0$ if and only if*

$$\boxed{a < \frac{\rho_0}{4\sqrt{2}L}}. \quad (18)$$

For a prescribed $q_0 < 1$, it is sufficient that

$$|\Delta\beta| < \frac{q_0\rho_0}{4\sqrt{2}L}.$$

Remark 9.3. The bound (18) is sufficient within this majorant method, not a claim about the sharp maximal flow time. A longer numerical segment may still be certifiable if a sharper interval Lipschitz estimate makes its linear radii polynomial negative. Otherwise the same numerical path is split into shorter mathematical segments.

Remark 9.4 (Why the segment polynomial is linear). A tempting second application of the scale-loss estimate does not give a bounded Lipschitz constant for $D\mathcal{F}$ in the same norm (17). Two path factors produce a d^{-2} integrand; Volterra integration leaves a $d^{-1/2}$ divergence after multiplication by the path weight. Consequently, the hypothesis of the abstract quadratic Newton–Kantorovich theorem is *not* automatic on $\mathfrak{E}_{a,\kappa}$. This revision therefore certifies each segment with the first-order contraction estimate already proved in Theorem 9.1. A quadratic polynomial may be used only if a separate, same-space interval theorem supplies its derivative bound.

Theorem 9.5 (Continuation and Cauchy endpoint). *Assume a solution exists on a maximal base interval and that one smaller strip $\rho_* > 0$, all admissibility margins, the SP/SP' majorants, and the scale-Lipschitz constants remain uniform. Then*

$$\sup_{\beta} \|\Psi_{\beta}[h_{\beta}]\|_{\rho_*} < \infty, \quad \|h_{\beta_2} - h_{\beta_1}\|_{\rho_*} \leq C_* |\beta_2 - \beta_1|.$$

Hence the endpoint exists in $\mathcal{A}_{\rho_,0}$ and the flow can be restarted. Premature breakdown can occur only if a certified margin tends to zero or a certified norm diverges.*

9.1 Volterra residual and the justified radii polynomial

For $u = h - h_a$, define

$$\mathcal{F}(u) = u - \mathcal{V}(\widehat{\Psi}[h_a + u]) \quad \text{on } \mathfrak{E}_{a,\kappa}. \quad (19)$$

Then

$$D\mathcal{F}(u) = I - \mathcal{V}D\widehat{\Psi}[h_a + u] \in \mathcal{L}(\mathfrak{E}_{a,\kappa}).$$

The strip loss is absorbed by the Volterra integral and the path-space weight.

Theorem 9.6 (Linear segment radii polynomial). *Let $\tilde{u} \in \mathfrak{E}_{a,\kappa}$ be a numerical path. Assume that the Volterra fixed-point map $\mathcal{T}$ is defined on $\overline{B}_r(\tilde{u})$, that this entire ball satisfies all scale-admissibility guards, and that interval estimates prove*

$$\|\mathcal{F}(\tilde{u})\|_{\mathfrak{E}} \leq Y, \quad \|\mathcal{T}(u) - \mathcal{T}(v)\|_{\mathfrak{E}} \leq q \|u - v\|_{\mathfrak{E}}, \quad q < 1.$$

Define

$$\boxed{p(r) = Y + (q - 1)r.} \quad (20)$$

If $p(r) < 0$, the segment contains a unique exact normalized solution in the ball and

$$\|u_* - \tilde{u}\|_{\mathfrak{E}} \leq \frac{Y}{1 - q} < r.$$

For every chosen endpoint strip $0 < \rho_{\text{end}} < \rho_0 - \kappa a$, its explicit endpoint error is

$$\varepsilon_{\text{path}} = \frac{r}{\sqrt{\rho_0 - \kappa a - \rho_{\text{end}}}}, \quad \|u_*(a) - \tilde{u}(a)\|_{\rho_{\text{end}}} < \varepsilon_{\text{path}}. \quad (21)$$

Proof. The inequality $p(r) < 0$ is exactly $Y + qr < r$, so $\mathcal{T}$ maps the ball into itself. Theorem 3.1 gives existence, uniqueness, and the stated error bound. $\square$

Theorem 9.7 (Validated segmented march). *Suppose the base interval is covered by finitely many re-anchored segments. If every segment has a strictly negative linear polynomial (20) in its own fixed path space, every radii ball is scale-admissible, and interval endpoint evaluation places its complete endpoint ball inside the admissible start ball of the next segment, then there exists a unique global anchor-pure march. Its final error is bounded by Theorem 4.4.*

10 Reconstruction and Kneser identification

Proposition 10.1 (Preservation of Abel structure). *A solution of the normalized hub flow preserves $T(0) = 1$, the tetration equation, real symmetry, and orientation as long as the admissibility margins remain positive.*

Proof. Normalization and the functional equation are built into the QH representation. Real symmetry is preserved by the RH operator, and the derivative lower bounds in (Z2)–(Z3) preserve orientation. $\square$

10.1 Stopped logarithmic peeling

Definition 10.2 (Stopped peeling certificate). An SP certificate consists of a finite logarithmic chain from the seed z_N to a fixed handoff level, interval lower bounds on all intermediate moduli, distances from logarithmic cuts, derivative bounds, and a summable tail estimate for the difference of consecutive approximants.

Proposition 10.3 (Existence of the raw branch). *An SP certificate implies locally uniform convergence of the peeled approximants to a holomorphic raw branch Q_β satisfying (8). The derivative sequence converges locally uniformly, and the truncation error is bounded by the stored tail majorant.*

Corollary 10.4 (QH identification). *If Q_β is the certified raw branch and $U_\beta = \text{id} + p_\beta$ is the normalized degree-one lift produced by the hub flow, then*

$$T_\beta^{\text{rec}} = Q_\beta \circ U_\beta$$

is a real-analytic normalized solution of the tetration equation on the certified channel.

Hypothesis 10.5 (Complex peeling gate). The stopped peeling chain, its derivative chain, and the QH images admit certified complex continuations on a common overlap domain, with positive cut and zero-free margins.

Theorem 10.6 (Complex QH continuation). *Under the complex peeling gate, $T_d^{\text{rec}} = Q_d \circ U_d$ extends holomorphically to the certified common domain and satisfies the functional equation there. Once Theorem 11.14 identifies one germ with the regular Kneser branch, equality continues along every common analytic continuation path.*

11 Finite certification framework

Let Π denote a finite Fourier–Lobatto projection.

Theorem 11.1 (Completely computable segment certificate). *Assume interval bounds are available for*

1. *the finite and analytic-tail Volterra residuals Y_N, Y_T ;*
2. *a scale-Lipschitz majorant $\bar{L}$ for the complete normalized field, assembled from the interval QH, RH, complex-tail, differentiation, and product estimates on the whole candidate ball;*
3. *positive interval margins for every scale-admissibility condition, for $\kappa a < \rho_0$, and for one selected endpoint strip $\rho_{\text{end}} < \rho_0 - \kappa a$, together with raw-normalization and representation errors $\varepsilon_q, \varepsilon_{\text{repr}} \geq 0$.*

Set

$$Y = Y_N + Y_T, \quad q = \frac{4\sqrt{2}\bar{L}}{\kappa}, \quad p(r) = Y + (q - 1)r.$$

If $q < 1$, the radius- r ball remains inside every stored margin, and $p(r) < 0$, the segment is rigorously certified by Theorem 9.6. Every quantity reduces to finitely many outward-rounded interval operations, validated Fourier/Chebyshev conversions, analytic tails, and the Volterra constant from Theorem 9.1. The complete endpoint error passed to the next segment is

$$\varepsilon_{\text{end}} = \varepsilon_{\text{path}} + \varepsilon_q + \varepsilon_{\text{repr}}.$$

Remark 11.2. A finite-block approximate inverse may still reduce the residual or sharpen an implementation's predictor. It does not enter the proof profile used by Theorem 11.1; the machine checker labels that profile `ovsyannikov-contraction-v1`.

Definition 11.3 (Anchor certificate). An anchor certificate $\mathcal{A} \setminus \lfloor (\sigma_0)$ consists of an interval Chebyshev representation of T_e on a complex neighborhood of $[0, 1]$, a remainder bound, a positive lower bound on $|T'_e|$, and positive cut margins for the finitely many transfers up to the stopping index.

Theorem 11.4 (Uniform K1 data along a certified march). *If the anchor certificate holds and all segments are certified by Theorem 11.1, then the reconstructed family has explicit uniform lower bounds for strip width, derivative, and distance from all relevant cuts on the whole base interval.*

Lemma 11.5 (Computable M_1). *For $G_\beta \in \mathcal{A}_{\rho_G}$ and $\rho' < \rho_G$,*

$$M_1 = \sup_{\beta, z \in S_{\rho'}} |\partial_x G_\beta(z)| \leq \sup_{\beta} \left[\sum_{|k| \leq K_m} 2\pi |k| \left| \widehat{G}_k(\beta) \right| e^{2\pi\rho'|k|} + \frac{2e^{-\pi(\rho_G - \rho')(K_m+1)}}{e(\rho_G - \rho')} \|G_\beta\|_{\rho_G} \right].$$

Proof. For $\delta = \rho_G - \rho' > 0$ and $n \geq K_m + 1$,

$$2\pi n e^{-2\pi\delta n} = (2\pi n e^{-\pi\delta n}) e^{-\pi\delta n} \leq \frac{2}{e\delta} e^{-\pi\delta(K_m+1)}.$$

Multiplying the ρ_G -weighted Fourier tail by this factor gives the stated bound. The finite head is evaluated directly. $\square$

Theorem 11.6 (Global error budget). *If complete segment endpoint errors $\varepsilon_j = \varepsilon_{\text{path},j} + \varepsilon_{q,j} + \varepsilon_{\text{repr},j}$ and derivative bounds $M_{1,j}$ are certified on one common retained strip, then*

$$\left\| p_{\beta_d}^{\text{num}} - p_{\beta_d}^{\text{exact}} \right\| \leq \sum_{j=1}^m \varepsilon_j \prod_{k=j+1}^m \exp(M_{1,k} |\Delta\beta_k|).$$

The corresponding direct-versus-split path gate is bounded by the sum of the two route budgets.

Theorem 11.7 (Hardy boundary-value equivalence). *For $w \in \mathcal{A}_\rho$, the following are equivalent:*

1. $w \in \mathcal{A}_\rho^+$;
2. w has a bounded holomorphic periodic extension to $\Im z > -\rho$;
3. the extension converges uniformly to $\widehat{w}_0$ as $\Im z \rightarrow +\infty$.

Moreover,

$$|w(x + iy) - \widehat{w}_0| \leq e^{-2\pi(y+\rho)} \|w\|_\rho.$$

Definition 11.8 (Segment-uniform upper end-gluing certificate). For a closed marched base segment I , an end certificate $\mathcal{E}_I^+$ consists of:

1. a finite outward-rounded cover $I = \bigcup_\nu I_\nu$ and, on every base box, a Koenigs inverse $\psi_\beta = \chi_\beta^{-1}$ on one common half-plane H_Y , with the uniform bound

$$|\psi_\beta(w) - L_\beta| \leq C_L e^{-a_L \Im w},$$

2. interval Hardy enclosures for $\vartheta_\beta \in \mathcal{A}_p^+$ such that $F_\beta(w) = w + \vartheta_\beta(w)$ obeys the box-uniform strict inequality $\sup_{I_\nu \times H_Y} |\vartheta'_\beta| \leq \eta_\nu < 1$, together with a certified common image half-plane;
3. a finite cover of every (β, z) overlap on which interval Newton, including parameter dependence, proves uniqueness for

$$F_\beta(\chi_\beta(y)) = z,$$

and encloses both the reconstructed and Koenigs-side values in the same uniqueness box.

4. compatibility on adjacent base boxes and, at a re-anchoring point, an interval intersection that contains the outgoing endpoint ball and the incoming normalized start ball. The stored inclusion is performed on the full coefficient-and-tail ball, not only at its midpoint.

Theorem 11.9 (Segment-uniform Hardy fixed-point gluing). *If $\mathcal{E}_I^+$ holds, then for every $\beta \in I$, T_β^{rec} has a unique holomorphic continuation to the upper end and*

$$T_\beta^{\text{rec}}(z) \longrightarrow L_\beta \quad (\Im z \rightarrow +\infty)$$

locally uniformly in $\Re z$ and uniformly for β in every stored base box. Adjacent segment continuations agree on their certified overlaps.

Proof. By Theorem 11.7, ϑ_β converges exponentially to its constant mode. The box bounds $\eta_\nu < 1$ make F_β injective and give a holomorphic inverse on the certified channel. Therefore $\psi_\beta(F_\beta^{-1}(z)) \rightarrow L_\beta$. The finite overlap gate and the identity theorem glue this continuation to T_β^{rec} . All constants are interval-uniform on each I_ν . The last item of Definition 11.8 and uniqueness on the overlap give agreement at every re-anchoring boundary. $\square$

Theorem 11.10 (Segment-uniform two-sided fixed-point limits). *Assume $\mathcal{E}_I^+$ and conjugation symmetry on the continued domain. Write $L_{1,\beta}$ and $L_{2,\beta} = \overline{L_{1,\beta}}$ for the upper and lower fixed points. Then, for every $\beta \in I$ and every real x in the continued domain,*

$$\lim_{y \rightarrow +\infty} T_\beta^{\text{rec}}(x + iy) = L_{1,\beta}, \quad T_\beta^{\text{rec}}(x + iy) \longrightarrow L_{2,\beta} \quad (y \rightarrow -\infty).$$

Both convergences are locally uniform in x , uniform for β in every stored base box, and compatible at every certified re-anchoring boundary.

Proof. The upper limit, its uniformity, and compatibility are Theorem 11.9. For $y < 0$, conjugation symmetry gives $T_\beta^{\text{rec}}(x + iy) = \overline{T_\beta^{\text{rec}}(x - iy)}$. Apply Theorem 11.9 as $-y \rightarrow +\infty$ and take conjugates. $\square$

Theorem 11.11 (Paulsen–Cowgill Proposition 2). *Let $b > e^{1/e}$, let $L_{1,b}$ and $L_{2,b}$ be respectively the upper and lower fixed points of $z \mapsto b^z$, and let F_1, F_2 be analytic on*

$$H_{-2} := \{z \in \mathbb{C} : \Re z > -2\}.$$

Assume, for $j = 1, 2$,

1. $F_j(z+1) = b^{F_j(z)}$ on H_{-2} ;
2. $F_j(0) = 1$;
3. for every $x > -2$,

$$\lim_{y \rightarrow +\infty} F_j(x+iy) = L_{1,b}, \quad \lim_{y \rightarrow -\infty} F_j(x+iy) = L_{2,b}.$$

Then $F_1 = F_2$ on H_{-2} .

Proof. This is Proposition 2 of Paulsen and Cowgill [8]. In particular, the comparison domain is the half-plane $\Re z > -2$ and both vertical fixed-point limits are hypotheses. Holomorphy on the larger cut plane and conjugation symmetry are useful certification mechanisms below, but are not substitutes for the lower limit in the proposition. $\square$

Definition 11.12 (Finite recursive cut-plane certificate). A cut-plane certificate $\mathcal{C}_I^{-2}$ for a marched base segment I contains:

1. interval holomorphic models of the reconstructed QH germ on a finite cover of a fundamental transition region, with positive derivative, zero-exclusion, logarithmic-cut, and image-overlap margins;
2. compatible right-translation boxes generated by $F(z+1) = b^{F(z)}$ and compatible left logarithmic boxes approaching the boundary cut $(-\infty, -2]$ from within $\mathbb{C}_{-2}$;
3. a finite collection of interval invariant templates whose induction covers the unbounded translated cells and the cells approaching the two sides of the cut; each induction step stores a strict cut-distance or sector margin;
4. the upper-end models from $\mathcal{E}_I^+$, their conjugate lower-end models, and interval identity checks on all overlaps;
5. an interval proof of $F(0) = 1$, conjugation symmetry, and the functional equation on the seed boxes. Hashes and directed-rounding metadata bind every template to the corresponding segment data.

Theorem 11.13 (Cut-plane continuation and static identification). *If $\mathcal{C}_I^{-2}$ and $\mathcal{E}_I^+$ pass all strict checks, then, for every $\beta \in I$, the reconstructed germ has a single-valued holomorphic continuation to $\mathbb{C}_{-2}$, satisfies the functional equation there, is conjugation-symmetric, is normalized by $T_\beta^{\text{rec}}(0) = 1$, and satisfies both limits*

$$\lim_{y \rightarrow +\infty} T_\beta^{\text{rec}}(x+iy) = L_{1,\beta}, \quad \lim_{y \rightarrow -\infty} T_\beta^{\text{rec}}(x+iy) = L_{2,\beta} \quad (x > -2).$$

Consequently,

$$T_\beta^{\text{rec}} = T_\beta^K \quad \text{on } \mathbb{C}_{-2}.$$

Proof. The finite transition cover supplies compatible starting germs. Right translation is entire; each certified left step uses a single logarithmic branch on a box separated from its cut. The invariant templates prove by induction that these boxes cover every cell of $\mathbb{C}_{-2}$, including arbitrary approach to either side of the deleted ray. Interval identity checks make neighboring germs equal on nonempty open overlaps. Since $\mathbb{C}_{-2}$ is simply connected, the monodromy theorem gives one single-valued holomorphic continuation. The algebraic equation, symmetry, and normalization propagate from the seed cover by the same overlaps. Theorem 11.10 supplies both vertical fixed-point limits. Restrict T_β^{rec} and the regular Kneser branch to $H_{-2} = \{\Re z > -2\}$. They now satisfy every hypothesis of Theorem 11.11, so they agree on H_{-2} . Both functions are holomorphic on the connected domain $\mathbb{C}_{-2}$. Since H_{-2} is a nonempty open subset of that domain, the identity theorem extends the equality from H_{-2} to all of $\mathbb{C}_{-2}$. $\square$

Theorem 11.14 (Pointwise Kneser identification along the march). *Assume the segment certificates, SP/SP' , the complex peeling gate, $\mathcal{E}_I^+$, and $\mathcal{C}_I^{-2}$ hold on every segment. Then*

$$T_\beta^{\text{rec}} = T_\beta^K$$

throughout the march interval. The C^1 base dependence follows from the constructed flow and need not be assumed for the Kneser family.

Proof. Apply Theorem 11.13 on every segment. The certified overlap identities make these static identifications agree at each re-anchoring boundary. The constructed Volterra flow supplies the asserted base regularity. $\square$

Theorem 11.15 (Master computer-assisted certification theorem). *Let a finite re-anchored march connect the certified anchor at e to $d > e^{1/e}$. Suppose that each segment record contains:*

1. *an anchor/ SP/SP' bundle and a complete interval RH record $\mathcal{I}_{\text{RH}}$;*
2. *strict interval admissibility margins on the full candidate ball;*
3. *numbers $Y_j, \bar{L}_j, q_j, r_j, \rho_{\text{end},j}, \varepsilon_{\text{path},j}, \varepsilon_{q,j}, \varepsilon_{\text{repr},j}, \varepsilon_j$ with $0 \leq q_j < 1$ and*

$$\begin{aligned} q_j &\geq \frac{4\sqrt{2}\bar{L}_j}{\kappa_j}, & p_j(r_j) &= Y_j + (q_j - 1)r_j < 0, \\ \varepsilon_{\text{path},j} &\geq \frac{r_j}{\sqrt{\rho_{0,j} - \kappa_j a_j - \rho_{\text{end},j}}}, & \varepsilon_j &\geq \varepsilon_{\text{path},j} + \varepsilon_{q,j} + \varepsilon_{\text{repr},j}. \end{aligned}$$

4. *an interval endpoint inclusion into the next segment, a segment-uniform upper certificate $\mathcal{E}_{I_j}^+$, and the two-sided fixed-point certificate of Theorem 11.10;*
5. *compatible cut-plane certificates $\mathcal{C}_{I_j}^{-2}$.*

Then the march defines the unique regular Kneser family on the covered base interval. At the target, the stored numerical center has a rigorous error bounded by the Gronwall sum of Theorem 11.6; every evaluated value inherits that radius after interval composition with the raw branch.

Proof. Theorem 7.6 validates the exact vector field. The negative linear polynomial and Theorem 9.6 certify each local path; interval endpoint inclusions and Theorem 9.7 glue the paths. Theorem 11.6 propagates their radii. Finally, Theorems 11.9, 11.13, and 11.11 identify the result with Kneser tetration. $\square$

12 Numerical experiments and blind validation

This section records computations; it does not alter the hypotheses of the analytic theorems.

Blinding protocol. “Blind” is used in the following operational sense. Before any target-base value was read, the construction, the internal gates, their declared tolerances and the predicted error level were fixed; the reference values were held in a sealed archive whose SHA-256 digest was recorded in the run log at that time; unsealing consisted of reading the archive and computing the discrepancy, with no subsequent change to the construction, the gates or the predictors. The target-base reference is an evaluation of the regular branch by the author’s high-precision Kneser engine, a fork of Levenstein’s `fatou.gp` in PARI/GP [6], which is the engine used throughout Papers I–IV; it is referred to below as the *external engine*. The run metadata of the reference evaluation – working precision, theta truncation, and the digest and timestamp of the sealed archive – are part of the public repository recorded under Data and code availability.

What the ring closure tests, and what it does not. Both architectures start from the same certified anchor T_e . An error in the anchor is therefore common mode: it is invisible to the mutual comparison of the two reconstructions and is tested only by the comparisons against the external engine. The mutual comparison does test the layers in which the two implementations differ, namely kernel sampling, evolution scheme, normalization, and re-anchoring; the independence claim below should be read with that split in mind. The anchor itself is controlled separately by the anchor certificate of Definition 11.3 and by the gates of Appendix A.

12.1 Independent 65-node reconstruction at base 2

The first complete anchor-pure reconstruction used the exact raw branch Q_2 , a Chebyshev interpolation of the RH-selected kernel family in the base coordinate, and a Picard–Chebyshev characteristic flow. The target base itself was not an interpolation node. Increasing the number of nested Lobatto nodes gave the convergence shown in Table 2.

Table 2: Blind convergence of the first anchor-pure reconstruction of T_2 .

Base nodes	hub digits	pointwise T_2 digits	comment
10	10.5	8.0	initial blind reconstruction
17	17.1	–	intermediate test
33	25.9	15.3	nested refinement
65	30.6	20.7	final flow

Here “hub digits” is the number of correct decimal digits of the periodic state p_β at the target, measured by self-convergence against the next finer node set; the pointwise column is the discrepancy of the reconstructed T_2 against the external engine on the test set. The pointwise entry at 17 nodes was not evaluated, since that run served only as an intermediate resolution test.

At 65 nodes, the worst blind discrepancy on the original test set was 3.93×10^{-21} , while the sealed normalization gate predicted 2.83×10^{-21} . A later 48-point off-grid test gave 4.0×10^{-21} . The final path-splitting gate was 1.2×10^{-23} , the roundtrip residual 8×10^{-51} , the Picard residual 8.5×10^{-43} , the spectral tail estimate 3.8×10^{-18} , and the functional-equation residual 5.5×10^{-76} . The dominant floor is consistent with the pointwise RH field accuracy and interpolation amplification, not with the raw logarithmic ladder.

12.2 Two-phase re-anchoring march

A second architecture recomputed RH fields inside each segment, used a low-accuracy stage followed by a high-accuracy stage, regularized Anderson acceleration, neutral-mode reprojection, and a generalized raw branch anchored at the previous reconstructed state. Its defining raw branch is

$$Q_b(x) = \log_b^{\circ m}(\alpha T_c(m+x) + c_0), \quad \alpha = \frac{\log c}{\log b}, \quad c_0 = \frac{\log \alpha}{\log b}.$$

Because T_c itself was obtained from T_e by earlier segments, the construction remains anchor-pure.

Table 3 collects the principal blind milestones. The error is the maximum on the stated test grid after the target reference was unsealed.

For $b = 2$, the ten-segment march consisted of seven regular segments and three refined edge subsegments. The final gates were

$$|T(0) - 1| = 1.2 \times 10^{-18}, \quad |T(1) - 2| = 2.2 \times 10^{-47}, \quad \inf T' = 0.889.$$

The blind comparison triangle was

Table 3: Blind validation of the re-anchoring architecture. “Sealed” indicates a preregistered target reference.

Target base	segments	worst residual scale	blind error	decimal digits
2.6045	1	3.1×10^{-12}	4.9×10^{-12}	11.72
2.5	2	8.8×10^{-10}	8.7×10^{-10}	9.46
2.45 (sealed)	3	7.6×10^{-12}	6.8×10^{-12}	11.56
2.3 (sealed)	4	2.5×10^{-9}	3.8×10^{-9}	8.78
2.0 (double reference)	10	7.7×10^{-12}	7.6×10^{-11}	10.42

comparison	7 standard points	48 off-grid points
re-anchoring march vs. external engine	7.0×10^{-11}	7.6×10^{-11}
re-anchoring march vs. 65-node flow	7.0×10^{-11}	7.6×10^{-11}
65-node flow vs. external engine	3.9×10^{-21}	4.0×10^{-21}

Thus two independently designed anchor-pure methods meet at the same target to the error level predicted by the less accurate march. The agreement is not a tautological roundtrip: the two methods use different kernel sampling, evolution, and re-anchoring architectures. It does not, however, test the common anchor, and the two comparisons against the external engine share that reference.

An available check that was not used. By Remark 8.9 the mean of the reconstructed hub at the target equals the mixed-base mean shift $\mu_{e,2} = m(2)$, which Papers III–IV determine to forty digits from the stopped peel ladder without evaluating a Kneser branch at base 2. That reference is generated by a mechanism entirely different from the external engine evaluation used above, and it is available at a precision comparable to the 30.6 hub digits reported in Section 12. Adding it as a gate would close the one remaining gap in the present validation design.

12.3 Empirical Ovsyannikov step-size test

The final unrefined edge segment, from approximately $b = 2.048$ to $b = 2$, was the only segment that reached both iteration limits. It stalled at a residual 4.08×10^{-8} and accounted for essentially the entire intermediate ring-closure error. The interval pre-harvest for Corollary 9.2 classified the original base-coordinate length 0.0458 as admissible in the interior-gain regime but inadmissible in the edge-gain regime.

The segment was then split into three subsegments of length $\Delta\beta = 0.0153$, inside the certainly admissible band. Starting from the same coarse endpoint, the three subsegments converged to residuals

$$1.25 \times 10^{-12}, \quad 1.23 \times 10^{-12}, \quad 2.58 \times 10^{-12}.$$

This is a genuine prediction–failure–repair test of the analytic step-size criterion: the theory identified the location and scale at which subdivision was needed before the final blind comparison was opened.

12.4 Path gate and residual-to-error behavior

A direct route and a route split into two half-segments differed by 1.65×10^{-12} , below the combined a priori budget 1.45×10^{-11} . Across the march, the estimated strip lower bound decreased from about 0.425 at the anchor to 0.25–0.28 near $b = 2$, while

$$M_1 \in [0.00436, 0.00625].$$

The corresponding Gronwall factors were close to 1.0002, explaining why segment errors accumulated nearly linearly. The floating-point/interval a priori budget 7.11×10^{-11} and the measured final error 7.6×10^{-11} differ by only a factor 1.07. This agreement validates the scale of the internal predictor, but the budget used buffered residuals rather than radii-polynomial segment radii and is therefore labeled a precertificate.

13 Discussion and open problems

The reconstruction method reveals a useful structural separation. The tetraton itself is nonperiodic and belongs naturally to a fundamental-domain space, whereas the correction that carries the base dependence is periodic. The Riemann–Hilbert problem selects the correct homogeneous gauge, and the raw-branch derivative removes the apparently nonperiodic part of the transport equation. The resulting base flow is therefore substantially simpler than a direct evolution of T_β .

The two blind reconstructions show that this structure is numerically effective. In particular, the normalization gate predicts the pointwise reconstruction error, the direct/split path test detects implementation defects that a roundtrip can miss, and the Ovsyannikov bound correctly identifies the difficult edge regime near base 2. The experiments also suggest that an atlas of reconstructed bases could turn base continuation into a practical high-precision algorithm.

Several problems remain open.

13.1 A complete interval Riemann–Hilbert map

A full computer-assisted proof requires a validated complex-tube bound for the graded Gauss–Legendre evaluation of the RH trace. The finite Fourier head and Hardy projection are straightforward once the trace is enclosed; the main analytic task is a uniform panelwise remainder bound that remains sharp near the logarithmic endpoints.

13.2 Certified anchor-pure raw branches

The stopped peeling construction must be enclosed together with its base derivative on the entire radii ball. This requires SP and SP' certificates that control logarithmic cuts, derivatives, and summable tails uniformly in both the base parameter and the periodic state. Such a result would connect the local a posteriori solver directly to the anchor-pure lineage rather than merely to a numerical center.

13.3 Restoring analytic width between segments

The fixed-scale Ovsyannikov theorem produces an endpoint estimate on a smaller strip. A global certified march therefore needs either a strict inclusion of each outgoing coefficient-and-tail ball into the next segment's stronger start ball, or a strip-restoration theorem adapted to re-anchoring. The refined edge audit shows that repeating a one-microcell estimate without this handoff is insufficient.

13.4 A full computer-assisted march

Once the preceding three ingredients are available, the same certificate profile can be applied to every segment from e to 2. The remaining tasks are finite: segment radii, endpoint inclusions, two-sided fixed-point gluing, recursive continuation on $\mathbb{C} \setminus (-\infty, -2]$, and an independent array-level replay. Their completion would upgrade the blind reconstruction to a computer-assisted identification with Kneser tetraton.

13.5 Algorithmic base atlases and complex paths

The present experiments follow a real path in $\beta = \log \log b$. A natural computational problem is to choose an optimal sequence of stored anchors and certified step sizes for a requested target base and precision. A more ambitious question is whether complex base paths can avoid narrow-strip regions or reveal nontrivial base monodromy. Such paths would require a careful global treatment of branch choices and path independence.

14 Conclusion

We have developed an anchor-pure reconstruction of regular tetration across bases. The method combines a peeled raw branch, a periodic hub correction, Riemann–Hilbert selection, and a re-anchored analytic base flow. The main identities are exact, the local flow and finite certification framework are rigorous under explicit interval obligations, and two independent blind computations reconstruct T_2 from T_e with errors consistent with their internal predictors.

The present result is therefore both theoretical and computational: it explains how a known tetration can generate neighboring bases and demonstrates the construction at high precision without target-base input. A complete interval proof of the full march remains a well-defined extension rather than an assumption of the results reported here.

A Toward a computer-assisted proof

This appendix records the interval components that were implemented beyond the blind validation experiments and states precisely which additional bounds are required for a complete computer-assisted proof.

A.1 Interval precertificates

Using declared buffers around the computed Fourier coefficients, segment residuals, and fixed-point data, interval arithmetic produced:

$$\rho_G \geq 0.249\text{--}0.425, \quad M_1 \in [0.00436, 0.00625],$$

global Gronwall a priori budget = $[0, 7.11 \times 10^{-11}]$.

An anchor precertificate obtained a Chebyshev strip parameter $\rho_{\text{cheb}} = 8.62$, a geometric neighborhood width $\sigma_0 = 2.13$, a tail bound 3.8×10^{-44} , and a sampled derivative lower bound $\inf T'_e = 1.0918$. A refined edge segment was subsequently recomputed with its full Lobatto trajectory, high-accuracy RH modes, analytic tails, and hash-bound metadata, allowing a direct audit of the local certification pipeline.

These quantities are precertificates rather than final interval certificates. In particular, the residual inflation used in the first harvest was slightly too small: the final ring error was 1.60 times the sum of raw segment residuals, whereas the provisional buffer was 1.5. This confirms that heuristic residual buffers are unsuitable for a theorem: a rigorous global estimate must use the certified radii r_j . The harvest also predates the corrected derivative-tail bound in Lemma 11.5 and the path-to-endpoint factor (21); its quoted M_1 and global budget must therefore be recomputed rather than promoted into the final bundle.

A.2 Audit of a refined edge candidate

The refined edge trajectory was computed on the exact base interval

$$\beta_a = \frac{7}{8} \log \log 2, \quad \beta_b = \frac{11}{12} \log \log 2.$$

The candidate archive contains all seven base-Lobatto states, 48-point state-Chebyshev models, 96 real trace samples, resolved RH and residual modes, graded 24-point-per-panel Hilbert data, stopped Koenigs models, analytic tails, raw re-anchoring transforms, and SHA-256 bindings. Candidate centers were supplied by a target-base engine, while the RH field was rebuilt from Lambert- W /Koenigs data. This is admissible for testing a local a-posteriori certificate, but it does not establish the incoming anchor-pure lineage.

An outward-rounded replay produced the scalar intervals

$$\begin{aligned} Y &= [2.89398623506807 \times 10^{-10} \pm 8.15 \times 10^{-25}], \\ L &= [1.361330588130968 \times 10^4 \pm 8.70 \times 10^{-12}], \\ q &= [0.1495393411962912 \pm 9.19 \times 10^{-17}], \\ r &= [4.2535568887144 \times 10^{-10} \pm 3.01 \times 10^{-24}], \\ p(r) &= [-7.234965587670 \times 10^{-11} \pm 2.51 \times 10^{-24}]. \end{aligned}$$

These values verify the algebraic contraction inequalities for the stored model under its declared primitive bounds. They are not by themselves a segment proof.

The audit also detected and repaired an implementation defect: a declared beta-series order of 22 had in fact been truncated at degree 9 by the default series cap. The repaired Taylor-model code sets and checks the series length explicitly. It independently derives the beta-Taylor remainders, the stopped Koenigs remainders for Γ and Γ' , and the tower-tail rule. Across all 6×96 primitive rows, the worst derived remainders were approximately

$$E_\Gamma = 1.70 \times 10^{-14}, \quad E_{\Gamma'} = 1.78 \times 10^{-11}, \quad E_{\text{tail}} = 1.14 \times 10^{-10}.$$

An independent primitive verifier regenerated these enclosures from the raw trajectory and found no containment violation at the tested panel points. Thus the formerly free beta, Koenigs, and tail budgets have been replaced by mathematically derived and replayed bounds.

Three proof obligations remain and prevent promotion to a segment certificate.

1. The graded Gauss-Legendre error formula is derived, but the required uniform complex-tube supremum of the Koenigs trace has not been evaluated by the two-variable interval model. Hence the old Hilbert remainder is not yet an accepted analytic bound.
2. The raw branch Q_β has not been regenerated from the incoming anchor by a complete SP/SP' interval bundle, and $T = Q_\beta \circ U$ has not been enclosed on the whole radii ball.
3. Most importantly, the replay certifies one very short microcell and then forms a nominal segment endpoint by summing 262144 copies. The microcell theorem returns an error in a weaker strip, whereas the next cell requires an input ball in the stronger start strip. No rigorous inclusion or strip-restoration theorem is supplied. The reported global factor $M_1 = 0.01$ cannot replace this missing analytic handoff because the normalized hub vector field loses one derivative on a fixed strip.

Accordingly, the mathematical status of this edge computation is:

independently replayed primitive precertificate, not a closed segment proof.

The computation nevertheless validates substantial low-level certificate machinery and isolates the remaining theorem-level obstruction.

A.3 Upper end-gluing experiment

The end-gluing machinery of Definition 11.8 was implemented independently of the target-base tetration engine and tested at the e anchor. The recorded gates were

$$|\chi(E(y)) - \chi(y) - 1| = 1.1 \times 10^{-79}, \quad \chi \circ \psi \text{ roundtrip} = 4.0 \times 10^{-37},$$

$$\vartheta\text{-periodicity} = 1.5 \times 10^{-30}, \quad \eta = \sup |\vartheta'| = 6.3 \times 10^{-4} < 1.$$

The negative Hardy modes were at most 6.9×10^{-7} at the tested height. This is consistent with the expected Hardy selection after amplification of the finite anchor-state noise. Eight overlap boxes passed the implemented uniqueness check, but the current overlap residual 5.8×10^{-3} is limited by Fourier truncation and by a heuristic uniqueness radius. Accordingly, the anchor experiment is evidence for the end-gluing mechanism, not yet the interval certificate $\mathcal{E}_e^+$ of Definition 11.8.

B Computational implementation and reproducibility

B.1 Software architecture and guarantee levels

The accompanying software design separates numerical construction, validation gates, and interval certification. A read-only atlas chooses the nearest admissible anchor; its historical base-2 and base-3 entries are explicitly tagged as target-sampled legacy data and are excluded from the certified anchor-pure path. A phase map represents $H_{c,b} = A_c \circ T_b$ and composes exactly according to

$$H_{e,b} = H_{e,c} \circ H_{c,b}.$$

No target-engine fallback is permitted.

1. **Fast.** A stored atlas map is evaluated directly. For an unstored base, a pluggable floating-point RH marcher creates a short local map and reports a forecast from tail, collocation, and conditioning terms.
2. **Validated.** The same center is accepted only after path-splitting, roundtrip, normalization, functional-equation, resolution, and residual-to-error gates pass at their declared tolerances.
3. **Certified.** Evaluation is enabled only after the certificate checker has accepted the exact schema and proof profile of Theorem 11.15. The result is a center-radius enclosure.

The reference implementation contains the mode controller, phase-map composition, validation gates, a precertificate auditor, tests, and the certificate schema. Research prototypes additionally generate and replay several interval primitives. Certified output is intentionally fail-closed until the complete interval RH map, SP/SP' raw-branch enclosure, segment handoff, and independent global replay are available.

B.2 Machine-readable certificate format

For each segment, a complete record should contain:

1. a, ρ_0, κ and the checks $\kappa a < \rho_0$ and $4\sqrt{2}L/\kappa < 1$;
2. the normalized candidate path, its Volterra residual, and finite/tail residual bounds Y_N, Y_T ;
3. $\bar{L}, q, Y, r, \rho_{\text{end}}, \varepsilon_{\text{path}}, \varepsilon_q, \varepsilon_{\text{repr}}, \varepsilon_{\text{end}}$, the directed interval evaluation of $p(r) = Y + (q-1)r < 0$, and (21);
4. interval enclosures of all Z1–Z7 margins on the full radii ball, plus SP/SP' and complex tail majorants;
5. M_1 , the segment Lipschitz factor, and the propagated global error budget;
6. full interval RH data $\mathcal{I}_{\text{RH}}$, endpoint inclusion, $\mathcal{E}_I^+$, and $\mathcal{C}_I^{-2}$;

7. the certified height domain, phase-to-value Lipschitz factor, raw-branch error, and floating midpoint-evaluation rounding budget;
8. the exact uniqueness reference Paulsen–Cowgill (2017), Proposition 2, its DOI [10.1007/s10444-017-9524-1](https://doi.org/10.1007/s10444-017-9524-1), the comparison domain $\Re z > -2$, both limits L_1, L_2 , the separate identity-theorem bridge to $\mathbb{C}_{-2}$, plus source, coefficient, and evaluated phase-map hashes;
9. an independent array-level replay trace which binds every accepted scalar and Boolean claim to outward-rounded source arrays.

A floating-point residual alone is not a certificate.

B.3 Numerical safeguards and reproducibility lessons

The following failures were detected by independent gates and are part of the reproducibility record.

1. A truncated tail guard removed a term of size approximately 6×10^{-4} ; a telescoping next-term bound repaired it.
2. A collocation transport form suffered spectral high-mode amplification; the characteristic formulation removed this CFL-type instability.
3. Inconsistent Fourier cutoffs leaked into the normalization mode; consistent filtering and neutral-mode reprojection repaired the drift.
4. An unregularized Anderson step produced coefficients of order 10^8 ; regularization, relaxation, a trust region, and history resets were necessary.
5. Evaluation below zero requires downward logarithmic peeling. Directly folding a Chebyshev state to the fractional part produced false order-one errors.
6. A path-splitting gate detected an integration-coefficient bug that a roundtrip gate could not see. Both gates are therefore required.
7. A declared beta-Taylor order of 22 was silently truncated to degree 9 by a global series cap. Certification code must set and assert every series length explicitly.
8. Repeating a one-microcell endpoint bound does not certify a long analytic-scale segment unless each outgoing weak-strip ball is rigorously included in the next strong-strip input ball, or a separate strip-restoration theorem is proved.

Data and code availability

The engine, data and scripts of the series are publicly available at <https://github.com/Lightrunnerwastaken/gp-tetration>. The supplementary material for this paper is part of that repository: the L^AT_EX source, the numerical tables, the reconstruction artifacts, the SHA-256 hash manifests, the sealed reference archives used for the blind comparisons of Section 12 together with their run metadata, and the research prototypes used for the interval audits. This manuscript is archived at <https://doi.org/10.5281/zenodo.21709404>. The certified software mode is intentionally fail-closed: supplementary precertificates do not enable a certified result unless the complete interval obligations of Theorem 11.15 are present and independently replayed.

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Authorship and AI Contribution Statement

This paper is Paper VI of a mixed-base tetration series produced in a directed human–AI collaboration, and the division of labor should be stated plainly. The author conceived the anchor-pure reconstruction program, built and maintains the high-precision tetration engine used for every computation in the series (a fork of Levenstein’s `fatou.gp` [6]; engine, data and scripts at <https://github.com/Lightrunnerwastaken/gp-tetration>), and directed the research: posing the questions, fixing the blinding protocol and the acceptance gates, setting priorities between rounds, auditing intermediate results, and deciding which claims met those gates.

Substantial parts of the mathematical development were produced by Anthropic’s Claude in a theory and proof role under that direction. This includes the normalized QH representation of Section 6, the exact raw-branch derivative identity and the tail cancellation of Theorems 8.2 and 8.5, the fixed-scale Ovsyannikov argument and its path space in Section 9, the finite certification framework of Section 11, and the identification of two defects that shaped the

final form of the paper: the failure of the quadratic Newton–Kantorovich profile in the path norm (Remark 9.4) and the missing strip-restoration step behind the microcell obstruction (Appendix A). An autonomous research agent operating under the same direction designed, implemented and executed the numerical pipeline, produced both reconstruction architectures and the interval replays, and detected the implementation defects recorded in Appendix B. OpenAI’s ChatGPT contributed to review and exposition, and to the original versions of Papers I and II of the series.

Every statement was audited by the author against the independent numerical gates described in Section 12 and Appendix A. In line with prevailing journal policies, AI systems are not listed as authors; the author assumes full and sole responsibility for the correctness of all statements and proofs, and readers should weigh the provenance described here in their assessment. The author’s role, in short, was closer to that of a research director than of a sole working mathematician; the author considers it a matter of scientific honesty that this be recorded rather than obscured.